

Horizon: Evidence-Conditioned Runtime Assurance for Autonomous Maritime Systems

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ABSTRACT

Horizon inserts an independent safety governor between a vessel’s decision-making AI and its actuators. It joins navigation, contact, platform, communication, perception, and neural-health evidence before passing, modifying, replacing, or rejecting each proposed command. A 60-episode development study compared five candidates across four encounter classes. No declared-recoverable branch collided and no unsafe or stale decision passed the independent gate. We select A5, an evidence-conditioned predictive and barrier-filter hybrid, for the hackathon prototype: it achieved 99.83% gate acceptance with three deadline misses. This is an engineering choice, not a certified ranking; all missions were censored, 40 branches exceeded an assumed uncertainty bound, and only three paired seeds were run per scenario.

CCS CONCEPTS

• **Computer systems organization** → **Robotic autonomy**; • **Computing methodologies** → *Computer vision*.

KEYWORDS

runtime assurance, autonomous maritime systems, safety governor, mechanistic interpretability, sparse autoencoder

1 SAFETY OBJECTIVE

Autonomous maritime systems can emit unsafe commands even when individual components appear healthy. Horizon therefore enforces one protected actuation path. The external AI proposes a command but cannot write to steering or propulsion. Assurance evaluates public observations and declared uncertainty; an exclusive gate revalidates identity, lineage, epoch, freshness, deadline, actuator limits, and final-command safety. A watchdog retains minimum-risk authority if supervision stops.

The governing rule is monotone: unknown, stale, invalid, unbounded, or late evidence may only preserve or reduce authority. Camera or neural evidence is a *perception-health input*, not proof of obstacle-free water.

2 CANDIDATE ARCHITECTURES

All candidates implement one bounded input-to-decision contract. A1 applies CPA/TCPA thresholds and Simplex switching. A2 uses an uncalibrated analytic Gaussian collision-risk approximation with Simplex recovery. A3 requires finite rollout and a complete recovery continuation. A4 searches a nonlinear vessel plant map for a barrier-compatible command and validates it with a full 3DOF rollout. A5 conditions bounds and speed on evidence health, attempts predictive recoverability, and invokes A4 when required.

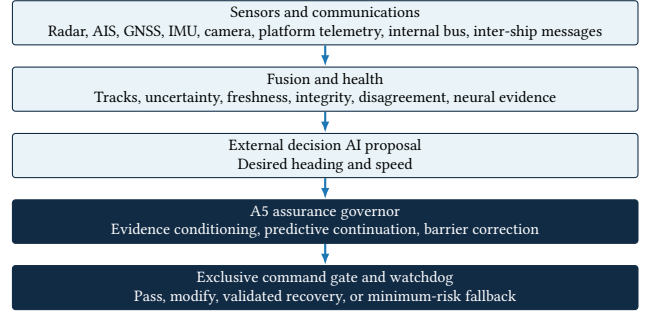


Figure 1: Protected Horizon control path. Evaluation truth is isolated from this path.

2.1 A5 decision path

Healthy evidence retains nominal bounds and a 6 m/s cap; degraded evidence expands bounds by 1.5 and caps speed at 3 m/s; invalid evidence caps speed at 1 m/s and forces recovery. A5 checks the request through a short handoff and complete recovery continuation. If that fails, it applies A4 and validates the correction with a complete rollout. Incomplete, infeasible, late, or unsupported results fall back to recovery or minimum-risk control.

3 EXPERIMENTAL METHOD

The matrix contained 12 paired identities: four scenarios, three seeds, and five candidates per identity, for 60 branches. Scenarios covered recoverable crossing, dense traffic, vessel degradation, and initially-unrecoverable encounters. Each branch ran for 30 simulated seconds under the idealized timing profile. Exact joins linked evaluation, diagnostics, and assumption audits; all collisions, timeouts, bound violations, and censored outcomes remained in their denominators [1].

4 RESULTS AND ARCHITECTURE CHOICE

All 15 branches in the deliberately unrecoverable stratum collided; none of the 45 declared-recoverable branches collided. No unsafe or stale result was accepted by the gate. A1 and A2 were computationally light but mostly passed proposals that the independent

Table 1: A1–A5 development results.

ID	Gate accepted	Deadline misses	Recoverable collisions	Unsafe/stale accepted
A1	8.87%	0	0	0
A2	8.73%	1	0	0
A3	99.66%	6	0	0
A4	99.44%	10	0	0
A5	99.83%	3	0	0

gate later rejected, giving acceptance rates below 9%. A3–A5 actively selected recovery or minimum-risk actions and exceeded 99.4% acceptance. A5 recorded the highest acceptance and fewer deadline misses than A3 or A4 (Fig. 2). We therefore recommend A5 for the hackathon integration while retaining A1 as the transparent baseline.

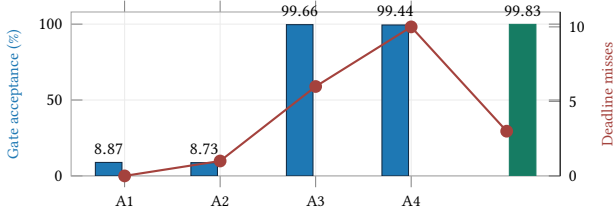


Figure 2: Gate acceptance (bars) and candidate deadline misses (red markers), 12 branches per candidate. Counts are descriptive, not a formal safety ranking.

5 NEURAL MECHANISTIC EVIDENCE

The implemented mechanistic-interpretability experiment is H4: a small sparse autoencoder (SAE) fitted to all 2,048 spatially pooled encoder channels from WaSR-T’s ResNet-101 layer4. Inputs are standardized and encoded into 16 nonnegative ReLU features. Runtime novelty would be the standardized reconstruction mean-square error plus an L_1 sparsity term. The 296-frame development fit reached its 3,000-epoch cap, reducing loss from 0.992552 to 0.168764 with zero dead features, but it did not meet the convergence criterion [2].

For exploratory causal testing, the largest adjacent-frame SAE-code change was selected independently for three predeclared frame pairs. Its decoder direction was injected into the real layer4 activation during six-frame temporal replay. The measured output was the absolute change in mean obstacle-class logit inside MODD2 obstacle boxes. Each selected direction exceeded both a random-direction and equal-norm noise control (Fig. 3), but the absolute effects were small and the selection was pair-specific.

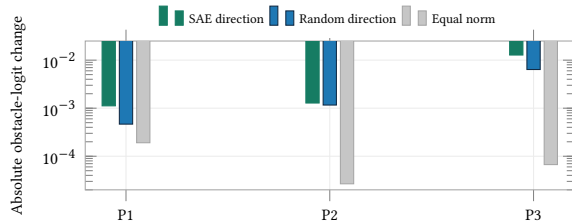


Figure 3: Offline H4 causal probes on three development pairs. Log scale.

H4 is *not deployed as a trusted runtime sensor*. Its claim gate remains closed because the SAE did not converge, the experiment used one sequence and one seed, no stable maritime semantic was assigned, and calibration and held-out partitions remain unopened. The live recorded-camera service currently defaults to H0 output entropy and permits H0/H1 only. H4 code therefore returns unknown unless a frozen causal-validation artifact explicitly passes

the claim gate. This fail-closed boundary prevents exploratory interpretation from silently increasing camera authority.

6 LIMITATIONS AND NEXT STUDY

The experiment cannot support certification or a final ranking. All 60 missions were censored, so route completion and cost are unknown. Forty branches exceeded an engineering uncertainty bound. No independent recovery boundary was supplied, front-end delay was idealized, and gate/watchdog actions affected trajectories. Three seeds per scenario are insufficient for statistical claims.

The next study should freeze methods before opening held-out data; run at least 1,000 paired episodes with 30 or more seeds per stochastic cell; separate preventable and unrecoverable encounters; validate perception thresholds on target CUDA hardware; and measure sensor-to-actuator latency. Deployment also requires qualified hydrodynamics, uncertainty coverage, communication-fault tests, and real ship interfaces.

7 CONCLUSION

Horizon demonstrates a coherent pattern: diverse ship and AI evidence informs a bounded governor while an independent gate owns actuation and fails closed. A5 is the best current hackathon architecture because it combines evidence-conditioned margins, predictive continuation, plant-aware correction, and the strongest measured acceptance/miss balance. SAE reconstruction and activation intervention remain exploratory until convergence, calibration, and held-out causal replication close the claim gate.

REFERENCES

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